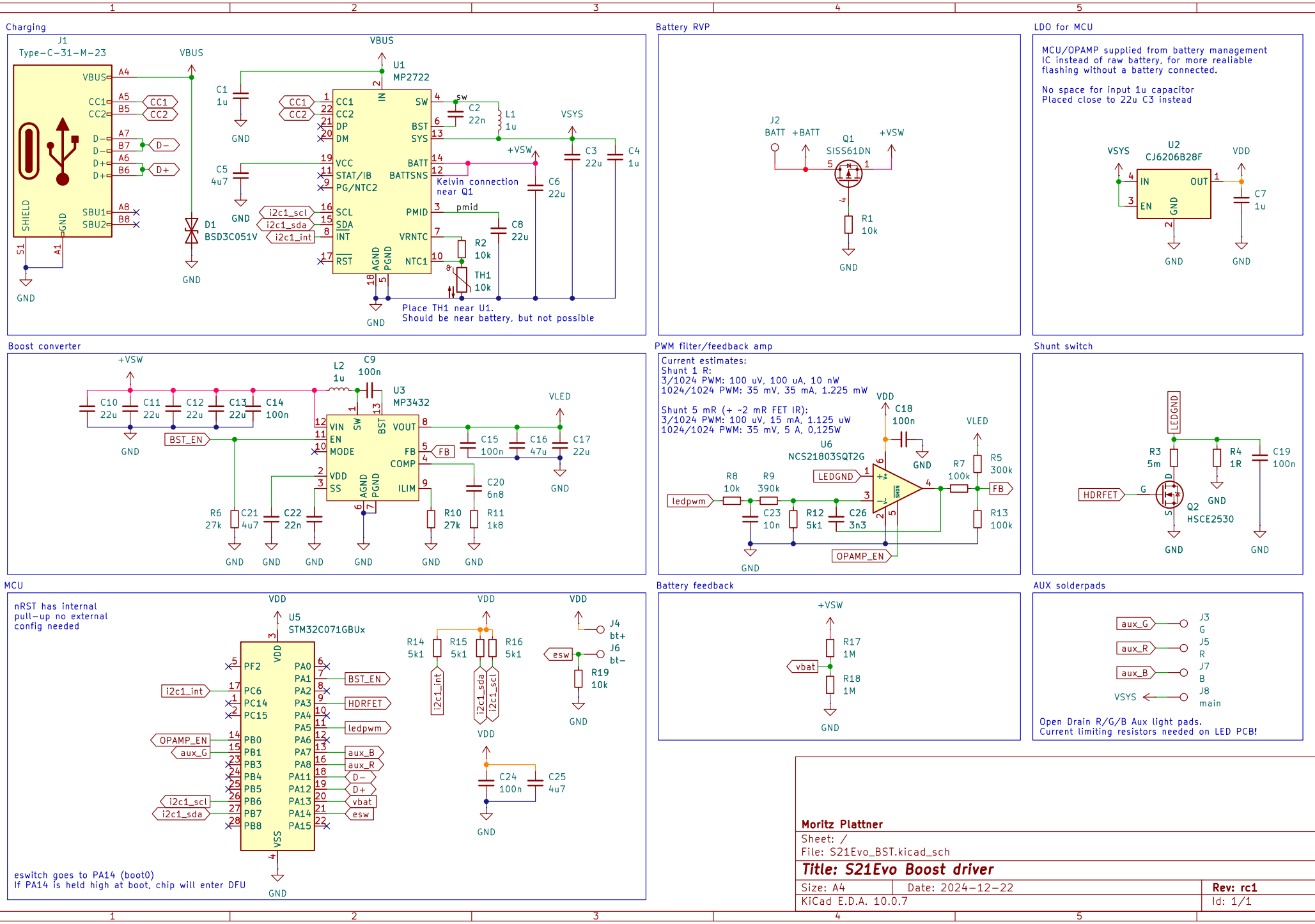

Charging
J1
Type-C-31-M-23
VBUS
U1
MP2722
Kelvin connection near Q1
Place TH1 near U1.
Should be near battery, but not possible
Battery RVP
Q1
SISS61DN
LDO for MCU
MCU/OPAMP supplied from battery management IC instead of raw battery, for more realiable flashing without a battery connected.
No space for input 1u capacitor
Placed close to 22u C3 instead
U2
CJ6206B28F
Boost converter
U3
MP3432
PWM filter/feedback amp
Current estimates:
Shunt 1 R:
3/1024 PWM: 100 uV, 100 uA, 10 nW
1024/1024 PWM: 35 mV, 35 mA, 1.225 mW
Shunt 5 mR (+ -2 mR FET IR):
3/1024 PWM: 100 uV, 15 mA, 1.125 uW
1024/1024 PWM: 35 mV, 5 A, 0,125W
U6
NCS21803SQT2G
Shunt switch
Q2
HSCE2530
MCU
nRST has internal pull-up no external config needed
U5
STM32C071GBUx
eswitch goes to PA14 (boot0)
If PA14 is held high at boot, chip will enter DFU
Battery feedback
AUX solderpads
Open Drain R/G/B Aux light pads.
Current limiting resistors needed on LED PCB!
Moritz Plattner
Sheet: /
File: S21Evo_BST.kicad_sch
Title: S21Evo Boost driver
Size: A4
Date: 2024-12-22
Rev: rc1
KiCad E.D.A. 10.0.7
Id: 1/1